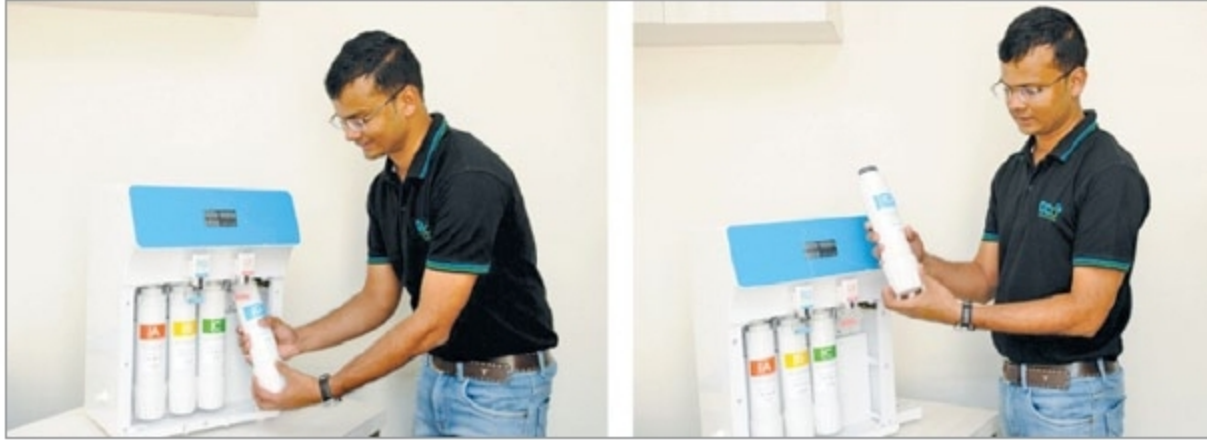




Quick and easy DIY filter change process demonstrated by Hasmukh Gulecha

water, for zero upfront or maintenance cost, and charges just ₹ 1 per litre of water consumed by the users.

Gulecha says, “Access to safe drinking water is a basic human right—not a privilege. Expenses for existing solutions, however, are very high. Moreover, water quality in India varies every 200 kilometres. Some regions have higher arsenic content in water, while others contain chloride, and some may contain iron and so on. The aim of OCEO is to purify water as per the specific problems existing in an area, as each contaminant requires a different approach for purification.”

OCEO is an Internet of Things (IoT)-enabled smart water purifier installed at the user’s location. The backbone of the technology lies in the blending of digital tools, Internet-linked sensors, online analytics and reports displayed on the devices. Water coming in from any source (tap, pump, tanker, etc) is first purified in the filter through six broad steps of filtration, carried out by a variety of elements. The six steps are polypro-

pylene filtration, activated carbon filtration, ultrafiltration (UF), reverse osmosis filtration, filtration through silver impregnated carbon filter and water mineralisation.

“When water is being dispensed, a sensor reads the quality of each drop of water and sends the data to a live cloud storage in real time,” informs Krishna. OCEO communicates with the network through GSM—a SIM card is fitted in the purifier. Data can be accessed as insightful analytics using OCEO’s mobile and Web applications. In addition, device health can also be verified using analytics. This facilitates predictive maintenance of the device, allowing auto-ordering of cartridges or filter membranes beforehand. The firm claims that the setup is simple enough for users to change filters and perform operations themselves (quick and easy DIY).

Users can consume any amount of water. Keeping in mind an Indian household consumption, OCEO machines are designed to dispense 15 litres of water every hour. OCEO has

kept the technology open source, and is inviting solution developers to join hands in developing new solutions for the increasing water challenges across the globe.

OCEO’s main target audiences are end consumers from cities as well as semi-urban and rural areas. The most interesting aspect of their offering is the total zero cost deployment model. This means customers do not have to pay anything for the device, installation or even maintenance (including filter membrane replacement). Instead, they have to pay as per their consumption, at the rate of ₹ 1 per litre.

The service must be availed on a prepaid basis. Users can purchase water credits through OCEO mobile application or website. For instance, they can purchase 200 water credits to consumer 200 litres of water. The predictive maintenance technology auto-orders fresh replacement membranes based on the incoming water quality and consumption data from the device.

Currently, the team is customising OCEO to fit the requirements of big offices. This includes increasing the water capacity, as offices require a lot more water daily as compared to households. This opens up more target customer groups for them. Dantewadiya says that, overall, OCEO plans to expand and cater to the consumers in India first, and once they meet the demands here, they will expand globally.

—Paromik Chakraborty, technical journalist, EFY

4 DIGITAL DAIRY FARMING, An Unexplored Land Of Infinite Possibilities

The dairy industry contributes around four per cent to India’s GDP, enjoying equal output value with rice and wheat. Still, gov-

ernment involvement is lacking to modernise and stabilise this sector. Farmers are losing money due to non-efficient dairy farming, comparatively

weak genetics of Indian cattle, high input cost and non-data-oriented farming. These bottlenecks are preventing Indian farmers to derive